

CE902 2024 Worked Solutions

Professional Practice and Research Methodology *Section B only (Questions 18-20)*

Scope note: the uploaded source file explicitly states “SECTION A IS NOT AVAILABLE”. Therefore this document solves all questions visible in the uploaded paper, namely Section B Questions 18 to 20, with full explanations and calculations.

Source paper	University of Essex CE902-7-SP (2024), uploaded by the user
Prepared as	Explanation-first worked answers with formulas and calculation steps
Coverage	Question 18, Question 19, and Question 20 only, because those are the questions present in the uploaded file

Question 18

(a) Recruitment criteria for the Software Engineer role

A sensible answer should be derived directly from the advert. The company is looking for someone who can contribute technically, work in a collaborative agile environment, and help deliver cloud-based industrial software. Four strong selection criteria are set out below, each linked to evidence in the role description.

Technical software development capability: The advert explicitly asks for software development experience, ideally in C#, ASP.NET Core, and Azure-based solutions. A strong candidate should therefore show sound programming ability, familiarity with modern software engineering practice, and some evidence of cloud application development.

Ability to design and implement complex solutions: The role is not limited to coding small isolated tasks. The engineer is expected to research, design, and implement technical solutions for industrial design data. That means the recruiter will value problem solving, system design thinking, and the ability to turn requirements into robust software components.

Teamwork and communication in an agile setting: The job requires collaboration with Product Owners, stakeholders, testers, product support teams, and the Scrum team. Therefore the successful candidate should be able to communicate clearly, participate in sprint planning and reviews, and work productively with colleagues from different roles and sites.

Professional attitude, curiosity, and willingness to learn and support others: The advert stresses curiosity, collaboration, mentoring, and a positive working environment. A recruiter

would therefore look for enthusiasm for learning, adaptability, responsibility, and the potential to coach or support less experienced colleagues over time.

(b) Structure of a competent application covering letter

A competent covering letter should be short, targeted, and organised so that the employer can quickly see why the applicant is suitable. A good structure is shown below.

1. Header	Applicant contact details, date, employer name, job title, and reference if one is provided.
2. Opening	A brief statement of the role being applied for and where it was advertised, followed by a professional expression of interest.
3. Main fit paragraph	A concise explanation of the applicant's relevant technical skills, qualifications, and experience, tailored to the employer's requirements.
4. Evidence paragraph	Specific examples showing achievements, projects, teamwork, communication, problem solving, or other qualities requested in the advert.
5. Motivation paragraph	Why the applicant wants the role and why the organisation appeals to them, showing genuine interest rather than generic praise.
6. Closing	A polite summary, availability for interview, thanks to the reader, and an appropriate formal sign-off.

In other words, the letter should not simply repeat the CV. Its job is to connect the applicant's background to the employer's needs and to make a persuasive case for interview.

Question 19

The paper gives the present value formula

$$PV = X / (1 + r)^t$$

For a lending decision, the net present value is then

$$NPV = PV(\text{inflow}) - \text{initial outflow}$$

Here the current outflow is £1,000, because that amount must be given to the cousin now.

(a) Compare the offer with an alternative return of 5% per annum

Future repayment promised after 2 years: £1,150

Discount rate for the comparison: 5% = 0.05

Step 1: Discount the future repayment back to present value: $PV = 1150 / (1.05)^2$

Since $(1.05)^2 = 1.1025$,

$$PV = 1150 / 1.1025 = 1043.08$$

Step 2: Compute net present value: $NPV = 1043.08 - 1000 = 43.08$

Because the NPV is positive (+£43.08), the promised repayment is worth more than the £1,000 that must be given up now when compared against a 5% opportunity cost. Therefore it is a financially worthwhile investment at 5%.

(b) Compare the same offer with an alternative return of 12% per annum

Now the appropriate discount rate is 12% = 0.12.

Step 1: $PV = 1150 / (1.12)^2$

Since $(1.12)^2 = 1.2544$,

$$PV = 1150 / 1.2544 = 916.77$$

Step 2: $NPV = 916.77 - 1000 = -83.23$

Here the NPV is negative (-£83.23), so the investment is not attractive if another opportunity really can earn 12% per year. In that case the better financial decision is to reject the loan and choose the alternative investment.

Question 20

The annual discount rate is 9%, so each cash flow at year t is converted to present value using

$$DCF_t = CF_t / (1.09)^t$$

The net present value is then the sum of all discounted cash flows, including the initial investment at year 0.

(a) Discounted cash flow calculations

Project A

Year	Cash flow	Discount expression	Discounted cash flow
0	\$-5.000 m	No discount at $t = 0$	\$-5.000 m
1	\$3.000 m	$3 / (1.09)^1$	\$2.752 m
2	\$2.000 m	$2 / (1.09)^2$	\$1.683 m
3	\$4.000 m	$4 / (1.09)^3$	\$3.089 m

4	\$3.000 m	$3 / (1.09)^4$	\$2.125 m
5	\$0.000 m	$0 / (1.09)^5$	\$0.000 m

Project B

Year	Cash flow	Discount expression	Discounted cash flow
0	\$-5.000 m	No discount at $t = 0$	\$-5.000 m
1	\$0.000 m	$0 / (1.09)^1$	\$0.000 m
2	\$0.000 m	$0 / (1.09)^2$	\$0.000 m
3	\$0.000 m	$0 / (1.09)^3$	\$0.000 m
4	\$0.000 m	$0 / (1.09)^4$	\$0.000 m
5	\$16.000 m	$16 / (1.09)^5$	\$10.399 m

(b) Net present value of each project

Project A: $NPV_A = -5 + 3/(1.09) + 2/(1.09)^2 + 4/(1.09)^3 + 3/(1.09)^4 + 0/(1.09)^5$

$NPV_A = 4.650$ million dollars

Project B: $NPV_B = -5 + 16/(1.09)^5$

$NPV_B = 5.399$ million dollars

Measure	Project A	Project B
NPV	\$4.650 m	\$5.399 m
Decision implication	Positive NPV	Higher positive NPV

(c) Recommendation

Assuming both projects carry the same risk, the financial analyst should recommend Project B. Both projects have positive NPVs, so both create value relative to the 9% discount rate. However, Project B has the larger NPV, so it adds more present-value wealth to the company. The fact that Project B pays later is already taken into account by discounting, so after adjusting for the time value of money, it still performs better than Project A.

Source file note: Section A was missing in the uploaded PDF, so this pack solves Section B only.